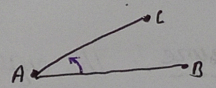
Arvind

Chapter ⇒ 2

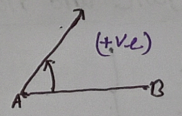
TRIGONOMETRY

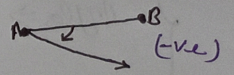
Trigonometry :- The science of measuring triangles is called trigonometry.

Angle :- An angle is the shape formed by the rotation of a ray around a point on itself while keeping it fixed



Sense of rotation :- When the direction of revolution is anti-clockwise than the angle is called positive and when revolution is clockwise the angle is called negative.





* Systems of measurement of angles

(1) Degree measure
(2) Grade measure
(3) Radian measure

(1) Degree measure :- If the rotation of the last arm from the starting arm is $\left(\frac{1}{360}\right)$ the fraction of a total revolution, we call the angle a measure of a degree, denoting it by 1°.

(i) 1 (Right angle) ⇒ 90°
(ii) 1 Degree ⇒ 60 minute, $60'$
(iii) 1 minute ⇒ 60 second, $60''$

⇒ The unit of measure of angle in this method is degree (1°).

Trigonometry

{Arvind}

1 minutes = 6°

1 hours = 30°

1 second = 6°

1 sec = $\frac{1}{60}$ min

1 min = $\left(\frac{1}{60}\right)^\circ$

hands hours make angle in 1 min $\Rightarrow \frac{360}{720}$

PBP / HHB

(1) $\tan\theta = \frac{\sin\theta}{\cos\theta}$ (2) $\sec\theta = \frac{1}{\cos\theta}$

(3) $\sin^2\theta + \cos^2\theta = 1$

(4) $\sin(A+B) = \sin A\cos B + \cos A\sin B$

(5) $\sin(A-B) = \sin A\cos B - \cos A\sin B$

(6) $\cos(A+B) = \cos A\cos B - \sin A\sin B$

(7) $\cos(A-B) = \cos A\cos B + \sin A\sin B$

(8) $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$

(9) $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}$

(10) $\cot(A+B) = \frac{\cot A\cot B - 1}{\cot A + \cot B}$

(11) $\cot(A-B) = \frac{\cot A\cot B + 1}{\cot B - \cot A}$

(12) $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B$
$= \cos^2 B - \cos^2 A$

(13) $\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B$
$= \cos^2 B - \sin^2 A$

Produt formula

(14) $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

(15) $2\cos A\sin B = \sin(A+B) - \sin(A-B)$

(16) $2\cos A\cos B = \cos(A+B) + \cos(A-B)$

(17) $2\sin A\sin B = \cos(A-B) - \cos(A+B)$

(18) $\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$

(19) $\sin C - \sin D = 2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$

(20) $\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$

(21) $\cos C - \cos D = 2\sin\frac{C+D}{2}\sin\frac{D-C}{2}$ $D > C$

To Express trigonometric Ratio of Angle 2A in terms of Angle A

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

(iii) $\cos 2A = 2\cos^2 A - 1$ or $1 + \cos 2A = 2\cos^2 A$

(iv) $\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$

To express $\sin 2A$ and $\cos 2A$ in terms of $\tan A$

(i) $\sin 2A = \frac{2\tan A}{1 + \tan^2 A}$

(ii) $\cos 2A = \frac{1 - \tan^2\theta}{1 + \tan^2\theta}$

(i) $\sin(3A) = 3\sin A - 4\sin^3 A$

(ii) $\cos(3A) = 4\cos^3 A - 3\cos A$

(iii) $\tan(3A) = \frac{2\tan A - \tan^3 A}{1 - 3\tan^2 A}$

(2.) Grade Measure :- In this method, the angle of 90° is divided into 100 equal parts, which is called grades.

g = grade

$90^\circ = 100^g$

1 Right angle ⇒ 100 grade or 100^g

1 Grade ⇒ 100 Minutes or $100'$

1 Minutes ⇒ 100 Second or $100''$

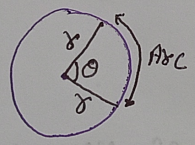
The unit of measure of angle in this Method is Grade (1^g)

(3) Radian measure :- The angle formed at the centre of the circle by arc of length equal to the radius in a circle is called a radian. 1 radian is expressed by 1^c

* Relation between angle, arc and radius

$$\text{Angle} \Rightarrow \frac{\text{Arc}}{\text{Radius}}$$

$$\theta = \frac{l}{r}$$



These value always in radian

$180^\circ \rightarrow \pi$ radian

$1^\circ \rightarrow \frac{\pi}{180}$ radian

$\{2\pi = 360^\circ\}$

⇒ Some usefull Points

* $60'$ M. $= 360$

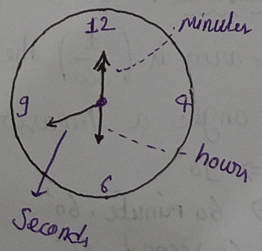
$1M = 6^\circ$

* $12h = 360$

$1h = 30^\circ$

* 60 Second $= 360$

$1\,Sec = 6^\circ$



* Radian measure ⇒ $\frac{\pi}{180} \times$ degree

* Degree measure ⇒ $\frac{180}{\pi} \times$ Radian

$1\,sec = \frac{1}{60}$ min

$1\,Min = \left(\frac{1}{60}\right)^\circ$

Q1 Convert 40°20' into radian measure.

$\{1^\circ = \frac{\pi}{180}$ radian$\}$

$1M = \left(\frac{1}{60}\right)^\circ$

$20'M =$? degree

$\frac{1}{60} \times 20$

$\left(\frac{1}{3}\right)^\circ$

40 degree + 20 minutes

$40 + \frac{1}{3}$

$\frac{120+1}{3}$

$\left(\frac{121}{3}\right)^\circ$

change radian $= \frac{\pi}{180} \times \frac{121}{3} \Rightarrow \frac{\pi 121}{540}$ Ans

Q2 Convert into 5°37'30" radian measure

$5^\circ + 37\,Min + 30\,Second$

$1\,sec = \frac{1}{60}$ Min

$30\,sec = \frac{1}{60} \times 30 = \frac{1}{2}$ Min

$1\,Min = \left(\frac{1}{60}\right)^\circ$

$\frac{75}{2}$ min ⇒ ? degree

$\frac{1}{60} \times \frac{75}{2}$

$\left(\frac{5}{8}\right)^\circ$

$37' + \frac{1}{2}'$

$\frac{74+1}{2}$

$\frac{75'}{2}$

$5^\circ + \frac{5}{8}^\circ$

$\frac{40+5}{8} = \left(\frac{45}{8}\right)^\circ$

change radian ⇒ $\frac{\pi}{180} \times \frac{45}{8} \Rightarrow \frac{\pi}{32}$ Ans

(3) Convert 6 radian into degree measure.

degree = $\frac{180}{\pi} \times$ radian

$\frac{180}{\pi} \times 6$

$= \frac{180 \times 6}{\frac{22}{7}} = \frac{90}{\frac{180 \times 6 \times 7}{22}}$

$\left(\frac{3780}{11}\right)^\circ$ Ans

$343\frac{7}{11} \Rightarrow 343^\circ + \left(\frac{7}{11}\right)^\circ$

1 Min $= \left(\frac{1}{60}\right)^\circ$

$343^\circ + \left(\frac{7}{11} \times 60\right)$ min

$343^\circ + \frac{420}{11}$ min

$343^\circ + 38\frac{2}{11}$ min

$343^\circ + 38' + \left(\frac{2}{11} \times 60\right)$ $\frac{120}{11} \approx 11'$

$343^\circ\ 38'\ 11''$ Ans

(4) Convert (-4) radians into degree measure.

radian into degree $\Rightarrow \frac{180}{\pi} \times$ radians

$\Rightarrow \frac{180}{\pi} \times 4$

$= \frac{720}{\pi} = \frac{360}{\frac{720 \times 7}{22}}$ $\quad \frac{22}{11}$

$= \frac{2520}{11} = 229\frac{1}{11}$

$229^\circ + \left(\frac{1}{11}\right)^\circ$

$229^\circ + \left(\frac{1}{11} \times 60\right)$ min

$229^\circ + \left(\frac{60}{11}\right)$ min

$229^\circ + \left(5\frac{5}{11}\right)$ min

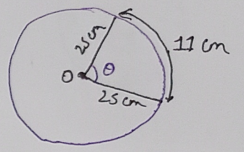
$229^\circ + \left(5 + \frac{5}{11}\right)$ min

$229^\circ + 5' + \left(\frac{5}{11} \times 60\right)$ sec

$229^\circ + 5' + 27.3''$

$-229^\circ\ 5'\ 27.3''$ Ans

Q5 Find in degree the angle subtended at the centre of a circle of radius 25 cm by an arc of length 11 cm.



Angle = $\frac{\text{Arc}}{\text{Radius}}$

Angle = $\frac{11}{25}$ radian

Change into degree = $\frac{36}{\frac{180}{\pi} \times \frac{11}{25}}$ $\quad 18$

$\frac{36 \times 11 \times 7}{22 \times 5}$ $\quad 2$

$\left(\frac{126}{5}\right)^\circ$ degree

$25\frac{1}{5}$

$25^\circ + \left(\frac{1}{5}\right)^\circ$

$25^\circ + \left(\frac{1}{5} \times 60^{12}\right)$ min

$25^\circ, 12'$ Ans

Q6 Find the length of an arc of a circle of radius 5 cm subtending a central angle measuring 45°.

$\theta = \frac{l}{r}$

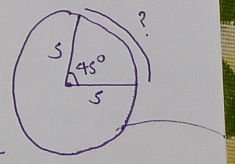
Change into radian $\Rightarrow \frac{45^\circ}{1} \times \frac{\pi}{180}$ $\quad 4$

$\frac{\pi}{4} = \frac{l}{5}$

$\frac{\pi}{4}$ radian

$\frac{5\pi}{4} = l$

$\frac{5 \times \frac{22}{7}}{4_2}$ $\quad \frac{55}{14}$ cm



* Relation Between Degree, Grade, Radian Measure

$$\text{Degree} \rightarrow \frac{D}{90} = \frac{2R}{\pi} = \frac{Gr}{100} \leftarrow \text{Grade}$$

(radian)

घड़ी की सुइयों से सम्बंधित प्रश्न

Q2 The minute hand of a watch is 15 cm long. How far does its tip move in 40 minutes? (use $\pi = 3.14$).

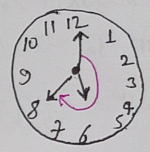
Given: $r = 1.5$

$\theta = \frac{l}{r}$

$\theta \rightarrow$ minute

$60m \rightarrow 360^\circ$

$1M = \frac{360}{60} \Rightarrow 6^\circ$

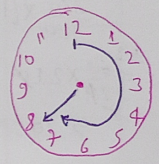


$40M \Rightarrow 40 \times 6 \Rightarrow 240^\circ$

Radian $= \frac{\pi}{180} \times 240 = \frac{4}{3}$

$\theta = \frac{4\pi}{3}$

$\frac{4\pi}{3}$



$\theta = \frac{l}{r}$

$\frac{4\pi}{3} = \frac{l}{1.5} \Rightarrow \frac{4 \times 3.14 \times 1.5}{3} = l$

$l = 4 \times 3.14 \times 0.5$

$l = 6.28$ cm

Q3 Find the angle between the minute hand of a clock and the hour hand when the time is 7:20 AM.

* Min $\rightarrow$ 20

1 min $\rightarrow 6^\circ$

20 min $\rightarrow 20 \times 6 \Rightarrow 120^\circ$

hours $\rightarrow$

$12h = 360^\circ$

$1h = \frac{360}{12} = 30^\circ$ change minute

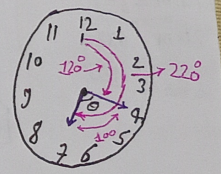
(12×60) min $= 360^\circ$

* $1M = \frac{360}{720}$

$\frac{360}{720}[7 \times 60 + 20]$

$\frac{360}{720}[420 + 20]$

$\frac{360}{720} \times 440 \Rightarrow 220$



अंतर $= 220 - 120$

100° Ans

Change into radian

$\frac{\pi}{180} \times 100$

$\frac{5\pi}{9}$ Ans

Q4 Find the degrees and radians the angle between the hour hand and the minute hand of a clock at half past three!

{3:30}

Sol:- 1M → 6°

तो 30M ⇒ 6×30 ⇒ 180°

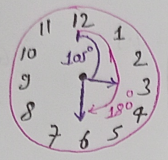
घण्टे वाली सुई सब minutes में = $\frac{360}{720}$

3:30 → $\frac{360}{720}(3\times60+30)$

$\frac{360}{720}\times(180+30)$

$\frac{360}{720}\times210$

⇒ 105



तथा 180 − 105

75°

Radian ⇒ $\frac{\pi}{180}\times75$

$\frac{5\pi}{12}$ Ans

Q5 Find the angle between the minute hand of a clock and the hour hand when the time is 12:20.

12:20

60M = 360°

1M = $\frac{360}{60}$ 6°

1M = 6°

20M = 6×20

= 120°

hours hand make angle in 1 minutes ⇒ $\frac{360}{720}$

12h ⇒ 360

1h = 30°

$\frac{360}{720}[0\times60+20]$

$\frac{360}{720}\times20$

1°

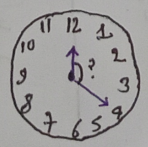
120° − 10°

110°

Change into radian

$\frac{\pi}{180}\times110$

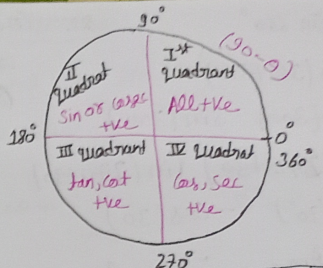
$\frac{11\pi}{18}$ Ans



T-Ratio

* Sin x · Cosec x
* Cos x · Sec x
* Tan x · Cot x

PBP / HHB (Cos x, Sin x, Tan x)



	0°	$\frac{\pi}{6}$ 30°	$\frac{\pi}{4}$ 45°	$\frac{\pi}{3}$ 60°	$\frac{\pi}{2}$ 90°	π 180°	$\frac{3\pi}{2}$ 270°	2π 360°
Sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	−1	0
Cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	−1	0	1
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not define	0	Not define	0

$\sin(90°-\theta)=\cos\theta$
$\cos(90°-\theta)=\sin\theta$
$\tan(90°-\theta)=\cot\theta$

$\sin(180°-\theta)=-\sin\theta$
$\tan(180°-\theta)=+\tan\theta$
$\cot(180°-\theta)=+\cot\theta$
$\sec(180°-\theta)=-\sec\theta$

When angle ... 180° and 360° than only sin -ve or +ve Change but or not ch... like: tan ... as no...

Ex $\sin(180°-\theta)$ ...

$\sin(270°-\theta)=-\cos\theta$
$\cos(270°-\theta)=-\sin\theta$
$\tan(270°-\theta)=+\cot\theta$
$\cot(270-\theta)=+\tan\theta$

$\sin(270°+\theta)=-\cos\theta$
$\cos(270°+\theta)=+\sin\theta$
$\tan(270°+\theta)=-\cot\theta$
$\sec(270°+\theta)=+\operatorname{cosec}\theta$

$\sin(360°-\theta)=-\sin\theta$
$\cos(360°-\theta)=+\cos\theta$
$\tan(360°-\theta)=-\tan\theta$
$\sec(360°-\theta)=+\sec\theta$

$\sin(360°+\theta)=+\sin\theta$
$\cos(360°+\theta)=+\cos\theta$
$\tan(360°+\theta)=\tan\theta$
$\sec(360°+\theta)=+\sec\theta$

Q1 $\sin 660^\circ$

$= \sin(360+300)$

$= \cos(300)\ \sin(300)$

$= \cos(270+30)\ \sin(270+30)$

$= -\sin(30^\circ) \quad -\cos(30)$

$= -1/2$ Ans $\quad -\sqrt{3}/2$ Ans

Q2 $\cos 1380^\circ$

$\cos(1080+300)$

$\cos(300)$

$\cos(270+30)$

$+\sin(30)$

$\sqrt{3}/2$ Ans

$1/2$ Ans

Q3 $\tan 1410^\circ$

$\tan(360^\circ \times 4 - 30^\circ)$ $\because \tan(360-\theta) = -\tan$

$\tan(30^\circ)$

$-\frac{1}{\sqrt{3}}$ Ans

Q4 $\operatorname{Cosec}(-2010^\circ)$

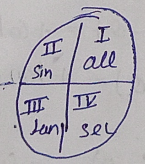
$-\operatorname{Cosec}(5 \times 360^\circ + 210)$ $\because \operatorname{Cosec}(360^\circ+210^\circ) = \sec(210$

$-\operatorname{Cosec}(210)$

$-\operatorname{Cosec}(270-60)$

$-(-\sec(60^\circ))$

$+2$



(3) Prove that

$$\frac{\cos(90^\circ+\theta)\sec(-\theta)\tan(180^\circ-\theta)}{\sec(360-\theta)\sin(180^\circ+\theta)\cot(90^\circ-\theta)} = -1$$

$$\frac{-\sin\theta \times \sec\theta \times -\tan\theta}{\sec\theta \times -\sin\theta \times \tan\theta} = -1$$

$-1 = -1$

Find the value of x in the following Equation:

(1) $\operatorname{Cosec}(90^\circ+\theta) + x\cos\theta\cot(90^\circ+\theta) = \sin(90^\circ+\theta)$

$\sec\theta + x\cos\theta(-\tan\theta) = \cos\theta$

$-x\cos\theta \times \frac{\sin\theta}{\cos\theta} = \cos\theta + \frac{1}{\cos\theta}$

$-x\sin\theta = \frac{\cos^2\theta - 1}{\cos\theta}$

$x\sin\theta = \frac{\sin^2\theta}{\cos\theta}$

$x = \tan\theta$

$\sin^2\theta + c^2 = 1$

$1 - \sin^2$

$\{\tan\theta = \frac{\sin\theta}{\cos\theta}\}$

$\{\sec\theta = \frac{1}{\cos\theta}\}$

$\{\sin^2\theta + \cos^2\theta = 1\}$

$\{\sin^2\theta = \cos^2\theta - 1\}$

Change

$\tan\theta = \cot\theta$

$\sin\theta = \cos\theta$

$\sec\theta = \operatorname{cosec}\theta$

(2) $\cos 24^\circ + \cos 55^\circ + \cos 125^\circ + \cos 204^\circ + \cos 300^\circ = \frac{1}{2}$

$\cos(24^\circ) + \cos(55^\circ) + \cos(180^\circ - 55^\circ) + \cos(180^\circ + 24^\circ) + \cos(360^\circ - 60^\circ) = \frac{1}{2}$

$\cos(24^\circ) + \cos(55^\circ) - \cos(55^\circ) - \cos(24^\circ) + \cos(60^\circ) = \frac{1}{2}$

$\cos(60^\circ) = \frac{1}{2}$

$\frac{1}{2} = \frac{1}{2}$

L.H.S = R.H.S

Hence Prove

(3) $\cos\frac{10\pi}{13} + \cos\frac{9\pi}{13} + \cos\frac{3\pi}{13} + \cos\frac{4\pi}{13} = 0$

sol:- $\cos\left[\pi - \frac{3\pi}{13}\right] + \cos\left[\pi - \frac{4\pi}{13}\right] + \cos\left[\pi - \frac{3}{13}\right] + \cos\frac{9\pi}{3} = 0$

$-\cos\frac{3\pi}{13} - \cos\frac{4\pi}{13} + \cos\frac{3\pi}{13} + \cos\frac{9\pi}{3} = 0$

$0 = 0$

(4) $\tan 5^\circ \tan 25^\circ \tan 45^\circ \tan 65^\circ \tan 85^\circ$

$[\tan 5^\circ \times \tan 85^\circ] \times \tan 45^\circ \times [\tan 25^\circ \times \tan 65^\circ]$

$[\tan(90-5) \times \tan 85] \times 1 \times [\tan(90-65) \times \tan 65^\circ]$

$\cot 5^\circ \times \tan 85^\circ \times 1 \times \cot 65^\circ \times \tan 65$

$1 \times 1 \times 1 = 1$

1 Ans

In any quadrilateral ABCD prove that :-

(1) $\sin(A+B) + \sin(C+D) = 0$

$\sin[360 - (C+D)] + \sin[360 - (A+B)]$

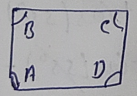
$(C+D) \to \theta$ $\quad$ θ

$-\sin(C+D) + \sin(C+D)$

$= 0$

$A+B+C+D = 360$

$A+B = 360 - (C+D)$



* If A, B, C, D are angles of a cyclic quadrilateral, prove that

(1) $\tan A + \tan B + \tan C + \tan D = 0$

$\tan(180-C) + \tan(180-D) + \tan C + \tan D = 0$

$-\tan C - \tan D + \tan C + \tan D = 0$

$0 = 0$

Hence prove

$A+B+C+D = 360^\circ$

$A+B = 180^\circ$ $\quad A =$

$A+C = 180^\circ$ $\quad A = 180 - C$

$B+D = 180^\circ$ $\quad B = 180 - D$

* Formulae for the Trigonometric Ratio of Sum and difference of two angles.

(1) $\sin(A+B) = \sin A\cos B + \cos A \sin B$

(2) $\sin(A-B) = \sin A\cos B - \cos A \sin B$

(3) $\cos(A+B) = \cos A\cos B - \sin A \sin B$

(4) $\cos(A-B) = \cos A\cos B + \sin A \sin B$

*(5) $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

*(6) $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

(7) $\cot(A+B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$

(8) $\cot(A-B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$

*(9) $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$

*(10) $\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$

Q1 Find the value of following

(i) $\sin 75°$

$\sin(45° + 30°)$

$\sin(A+B) = \sin A\cos B + \cos A \sin B$

$\sin 45° \cos 30° + \cos 45° \sin 30°$

$\frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2}$

$\frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}$

$\frac{\sqrt{3}+1}{2\sqrt{2}}$ Ans

Q2 if $\cos A = \frac{4}{5}$ & $\cos B = \frac{12}{13}$ $\frac{12}{13}, \frac{3\pi}{2} < A, B < 2\pi$ then find the value of the following

(i) $\cos(A+B)$

$\cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B$

$\sin A = \sqrt{1-\cos^2 A}$

$\sqrt{1-\frac{4^2}{5^2}}$

$\sqrt{1-\frac{16}{25}}$

$\frac{-3}{5}$

$\sin^2 A + \cos^2 A = 1$

$\sin A = \sqrt{1-\cos^2 A}$

90 / 180 / 0 / 270

IV Cos/Sec

Sin is (-ve)

$\sin B = \sqrt{1-\cos^2 B}$

$\sin B = \sqrt{1-\frac{12^2}{13^2}}$

$\sin B = \sqrt{1-\frac{144}{169}}$

$= \frac{-5}{13}$

$\cos\frac{4}{5} \times \cos\frac{12}{13} - \sin\left(\frac{-3}{5}\right) \times \sin\left(\frac{-5}{13}\right)$

$\left(\frac{4}{5} \times \frac{12}{13}\right) - \left(\frac{-3}{5} \times \frac{-5}{13}\right)$

$\frac{48}{65} - \frac{15}{65}$

$\frac{33}{65}$ Ans

Q2 If $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$, where $0 < A, B < \frac{\pi}{2}$, find the value of the following.

(i) $\sin(A+B)$

$\sin(A+B) = \sin A\cos B + \cos A \sin B$

$\sin A = \frac{4}{5}$ $\cos B = \frac{5}{13}$

$\sin B = \frac{12}{13}$ $\cos A = \frac{3}{5}$

$\frac{4}{5} \times \frac{5}{13} + \frac{3}{5} \times \frac{12}{13}$

$\frac{20}{65} + \frac{36}{65}$

$\frac{56}{65}$ Ans

$\sin^2 B + \cos^2 B = 1$

$\sin^2 B = 1 - \cos^2 B$

$\sin B = \sqrt{1-\cos^2 B}$

$= \sqrt{1-\left(\frac{5}{13}\right)^2}$

$= \sqrt{1-\frac{25}{169}}$

$= \sqrt{\frac{144}{169}}$

$= \frac{12}{13}$

$\sin^2 A + \cos^2 A = 1$

$\cos^2 A = 1 - \sin^2 A$

$\cos A = \sqrt{1-\sin^2 A}$

$\cos A = \sqrt{1-\left(\frac{4}{5}\right)^2}$

$= \sqrt{1-\frac{16}{25}}$

$= \sqrt{\frac{9}{25}}$

$= \frac{3}{5}$

Q3 If $\tan A = \frac{3}{4}$, $\cos B = \frac{9}{41}$, where $\pi < A < \frac{3\pi}{2}$ and $0 < B < \frac{\pi}{2}$, find $\tan(A+B)$.

PBP HHB

Sol: $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$

$\tan A = \frac{3}{4}$

$\tan B = \frac{40}{9}$

$\tan B = \frac{40}{9}$

$\cos B = \frac{9}{41}$

$\tan B = ?$

40, 41, ? = 9

$41^2 = 9^2 + 12^2$

$z = \sqrt{41^2 - 9^2}$

$z = \sqrt{1681 - 81}$

$z = \sqrt{1600}$

$z = 40$

$\frac{\frac{3}{4} + \frac{40}{9}}{1 - \frac{3}{4} \times \frac{40}{9}} = \frac{\frac{27+160}{36}}{1 - \frac{10}{3}}$

$\frac{187}{36} = \frac{187}{36} \times \frac{-7}{7}$

$\frac{-7}{3}$ 12

$-\frac{187}{84}$ Ans

$\frac{\pi}{2}$ / π / 0 / $\frac{3\pi}{2}$

tan B +ve

tan A +ve

Q4 if $\cos\alpha = \frac{1}{7}$ and $\cos\beta = \frac{13}{14}$ where α, β are acute angles, show that $(\alpha - \beta) = \frac{\pi}{3}$.

sol:- $\alpha - \beta = \frac{\pi}{3}$

$\cos(\alpha-\beta) = (\cos\alpha \cdot \cos\beta) + (\sin\alpha \cdot \sin\beta)$

$\cos(\alpha-\beta) = \left(\frac{1}{7} \times \frac{13}{14}\right) + \frac{\sqrt{48}}{7} \times \frac{\sqrt{27}}{14}$

$\frac{13}{98} + \frac{4\sqrt{3} \times 3\sqrt{3}}{98}$

$\frac{13}{98} + \frac{12 \times 3}{98}$

$\frac{13}{98} + \frac{36}{98}$

$\frac{49}{98} \quad \frac{1}{2}$

$\cos(\alpha-\beta) = \frac{1}{2}$

$\cos(\alpha-\beta) = \cos 60^\circ$

$\cos(\alpha-\beta) = \cos\frac{\pi}{3}$

$(\alpha-\beta) = \frac{\pi}{3}$

Hence Proved"

$\sin^2\alpha + \cos^2\beta = 1$

$\sin\alpha = \sqrt{1-\cos^2\beta}$

$\sin\alpha = \sqrt{1-\left(\frac{1}{7}\right)^2}$

$\sin = \sqrt{1-\frac{1}{49}}$

$\sin\alpha = \frac{\sqrt{48}}{7} = \frac{4\sqrt{3}}{7}$

$\sin^2\beta + \cos^2\beta = 1$

$\sin\beta = \sqrt{1-\cos^2\beta}$

$\sin\beta = \sqrt{1-\left(\frac{13}{14}\right)^2}$

$\sin\beta = \sqrt{1-\frac{169}{196}}$

$\sin\beta = \sqrt{\frac{196-169}{196}}$

$= \sqrt{\frac{27}{196}}$

$\frac{\sqrt{27}}{14} = \frac{3\sqrt{3}}{14}$

Q4 $\sin(n+1)A \sin(n+2)A + \cos(n+1)A \cos(n+2)A = \cos A$

$\cos(\underbrace{nA+A}_{A}) \cdot \cos(\underbrace{nA+2A}_{A}) + \sin(\underbrace{nA+A}_{A}) \cdot \sin(\underbrace{nA+2A}_{A})$

$\because \cos(A-B)$

$\Rightarrow \cos[nA+A-(nA+2A)]$

$\cos[nA+A-nA-2A]$

$\cos(-A)$

$\cos A = \cos A$

Q5 Prove that $\sin^2\left(\frac{\pi}{8}+\frac{A}{2}\right) - \sin^2\left(\frac{\pi}{8}-\frac{A}{2}\right) = \frac{1}{\sqrt{2}}\sin A$

$\sin^2 a - \sin^2 b$ using:- $\sin\left(\frac{\pi}{8}+\frac{A}{2}+\frac{\pi}{8}-\frac{A}{2}\right) \times \sin\left(\frac{\pi}{8}+\frac{A}{2}+\frac{A}{2}-\frac{\pi}{8}\right) =$

$\sin\left(\frac{2\pi}{8}\right) \times \sin\left(\frac{2A}{2}\right) = \frac{1}{\sqrt{2}}\sin A$

$\sin\frac{\pi}{4} \times \sin A = \frac{1}{\sqrt{2}}\sin A$

$\sin 45^\circ \times \sin A = \frac{1}{\sqrt{2}}\sin A$

$\frac{1}{\sqrt{2}} \times \sin A = \frac{1}{\sqrt{2}}\sin A$

$\frac{1}{\sqrt{2}}\sin A = \frac{1}{\sqrt{2}}\sin A$ Any

Hence proved

* Product formulae

(i) $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

(ii) $2\cos A\sin B = \sin(A+B) - \sin(A-B)$

(iii) $2\cos A\cos B = \cos(A+B) + \cos(A-B)$

(iv) $2\sin A\sin B = \cos(A-B) - \cos(A+B)$

[Transformation of sum or difference into product]

(i) $\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$

(ii) $\sin C - \sin D = 2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$

(iii) $\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$

(iv) $\cos C - \cos D = 2\sin\frac{C+D}{2}\sin\frac{D-C}{2}$. if $D > C$

Qi $\cos 75^\circ \cos 15^\circ$

$\frac{1}{2} \times 2 \times \cos 75^\circ \cos 15^\circ$ making in the form of formula.

$\because 2\cos A\cos B = \cos(A+B) + \cos(A-B)$ using formula

$\frac{1}{2}[\cos 90^\circ + \cos 60^\circ]$

$\frac{1}{2}[0 + \frac{1}{2}]$

$\frac{1}{4}$ Ans

(ii) $\cos 22\frac{1}{2}^\circ \sin 67\frac{1}{2}^\circ$

$\because 2\cos A \cdot \sin B = \sin(A+B) - \sin(A-B)$ using formula

$\frac{1}{2} \times 2\cos 22\frac{1}{2}^\circ \sin 67\frac{1}{2}^\circ$

$\frac{1}{2} \times 2[\sin 22\frac{1}{2} + 67\frac{1}{2} - \sin(22\frac{1}{2} - 67\frac{1}{2})]$

$\frac{1}{2}[\sin\frac{45}{2} + \frac{135}{2}] - \sin(\frac{45}{2} - \frac{135}{2})]$

$\frac{1}{2}[\sin\frac{180}{2}] - \sin(\frac{-90}{2})$

$\frac{1}{2}(\sin 90^\circ) - \sin(-45^\circ)$

$\frac{1}{2}[1 + \frac{1}{\sqrt{2}}]$

$\frac{1}{2}[\frac{\sqrt{2}+1}{\sqrt{2}}]$

$\frac{\sqrt{2}+1}{2\sqrt{2}}$

Q3 Prove that

$\sin 10^\circ \sin 50^\circ \sin 60^\circ \sin 70^\circ = \frac{\sqrt{3}}{16}$

$\sin 60^\circ \times \underbrace{\sin 10^\circ \times \sin 50^\circ}_{A \quad B} \sin 70^\circ = \frac{\sqrt{3}}{16}$

$\frac{\sqrt{3}}{2} \times \frac{1}{2} \times 2 \times \sin 10 \times \sin 50 \sin 70^\circ = \frac{\sqrt{3}}{16}$ ← $\boxed{\sin A \sin B}$ using

$\frac{\sqrt{3}}{2} \times \frac{1}{2} [\cos(10-50) - \cos(10+50)] \times \sin 70^\circ = \frac{\sqrt{3}}{16}$

$\frac{\sqrt{3}}{2} \times \frac{1}{2} [\cos(-40) - \cos(60)] \times \sin 70^\circ = \frac{\sqrt{3}}{16}$

$\frac{\sqrt{3}}{4} [\cos(40) - \cos(60)] \times \sin 70^\circ = \frac{\sqrt{3}}{16}$

$\frac{\sqrt{3}}{4} [\sin 70^\circ \cdot \cos 40 - \sin 70^\circ \cdot \cos 60^\circ] = \frac{\sqrt{3}}{16}$

$\frac{\sqrt{3}}{4} [\sin 70^\circ \cos 40^\circ - \frac{1}{2} \sin 70^\circ] = \frac{\sqrt{3}}{16}$

$\frac{\sqrt{3}}{4} \times \frac{1}{2} [2 \sin 70^\circ \cos 40^\circ - 1 \cdot \sin 70^\circ] = \frac{\sqrt{3}}{16}$ ← (using formula)

$\frac{\sqrt{3}}{8} [\sin(110) + \sin 30^\circ) - \sin 70^\circ] = \frac{\sqrt{3}}{16}$

$\frac{\sqrt{3}}{8} [\sin(180-70) + \sin 30^\circ - \sin 70^\circ = \frac{\sqrt{3}}{16}$

$\frac{\sqrt{3}}{8} [\sin 70 + \sin 30^\circ - \sin 70^\circ] = \frac{\sqrt{3}}{16}$

$\frac{\sqrt{3}}{8} \times \frac{1}{2} = \frac{\sqrt{3}}{16}$

$\frac{\sqrt{3}}{16} = \frac{\sqrt{3}}{16}$

Hence Proved

Express each of the following as a product.

(i) $\cos 4\theta + \cos 8\theta$

$\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right) \cdot \cos\left(\frac{C-D}{2}\right)$

$= 2\cos\left(\frac{4\theta+8\theta}{2}\right) \cos\left(\frac{4\theta-8\theta}{2}\right)$

$= 2\cos\left(\frac{12\theta}{2}\right) \cos\left(\frac{-4\theta}{2}\right)$

$= 2\cos 6\theta \cdot \cos(-2\theta)$ $\{\cos(-\theta) = +\cos\theta\}$

$= 2\cos 6\theta \cdot \cos 2\theta$

(ii) $\cos 6\theta - \cos 8\theta$

$\cos(C) - \cos(D) = -2\sin\left(\frac{C+D}{2}\right) \cdot \sin\left(\frac{C-D}{2}\right)$

$= -2\sin\left(\frac{14\theta}{2}\right) \cdot \sin\left(\frac{-2\theta}{2}\right)$

$-2\sin 7\theta \times -\sin\theta$

$+ 2\sin 7\theta \cdot \sin\theta$

Prove that

(1) $\frac{\cos 4A + \cos 3A + \cos 2A}{\sin 4A + \sin 3A + \sin 2A} = \cot 3A$

$\frac{\cos 4A + \cos 2A + \cos 3A}{\sin 4A + \sin 2A + \sin 3A} = \cot 3A$

$\frac{2\cos(3A)\times\cos(A) + \cos 3A}{2\sin(3A)\times\cos(A) + \sin(3A)} = \cot 3A$

$\frac{\cos(3A)[2\cos A + 1]}{\sin(3A)[2\cos A + 1]} =$ $\cot(A) = \frac{\cos A}{\sin A}$

$\frac{\cos(3A)}{\sin(3A)} = \cot 3A$

$\cot 3A = \cot 3A$

(2) $\cos 55^\circ + \cos 65^\circ + \cos 175^\circ = 0$

$\cos(55) + 2\cos\left(\frac{65+175}{2}\right)\times\cos\left(\frac{65-175}{2}\right) = 0$

$\cos 55^\circ + 2\cos(120)\cos(-55) = 0$

$\cos 55^\circ + 2\cos 120^\circ \cos 55^\circ = 0$

$\cos 55 + 2\cos(90+30)\cos(55) = 0$

$\cos 55^\circ + 2\times -\sin(30)\times\cos 55^\circ = 0$

$\cos 55^\circ + 2\times -\frac{1}{2}\cos 55^\circ = 0$

$\cos 55^\circ - \cos 55^\circ = 0$

$0 = 0$

"Hence proved!!"

To Express trigonometric Ratios of Angle 2A in terms of Angle A.

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

(iii) $\cos 2A = 2\cos^2 A - 1$ or $1 + \cos 2A = 2\cos^2 A$

(iv) $\cos 2A = 1 - 2\sin^2 A$ or $1 - \cos 2A = 2\sin^2 A$

(v) $\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$

To express $\sin 2A$ and $\cos 2A$ in terms of $\tan A$

(1) $\sin(2A) = \frac{2\tan A}{1 + \tan^2 A}$

(2) $\cos 2A = \frac{1 - \tan^2\theta}{1 + \tan^2\theta}$

(To express trigonometric ratio angle 3A in terms of angle A)

(1) $\sin(3A) = 3\sin A - 4\sin^3 A$

(2) $\cos(3A) = 4\cos^3 A - 3\cos A$

(3) $\tan(3A) = \frac{2\tan A - \tan^3 A}{1 - 3\tan^2 A}$

(1) If $\tan\theta = \frac{b}{a}$ then find the value of $a\cos 2\theta + b\sin 2\theta$

* $\sin 2A = \frac{2\tan A}{1+\tan^2 A}$

* $\cos 2A = \frac{1-\tan^2\theta}{1+\tan^2\theta}$

$$\frac{a \times 1-\tan^2\theta}{1+\tan^2\theta} + \frac{b \times 2\tan\theta}{1+\tan^2\theta}$$

$$a \times \left[\frac{1-(\tan\theta)^2}{1+(\tan\theta)^2}\right] + b \times \left[\frac{2\tan\theta}{1+(\tan\theta)^2}\right]$$

$$a \times \left[\frac{1-\frac{b^2}{a^2}}{1+\frac{b^2}{a^2}}\right] + b \times \left[\frac{2 \times \frac{b}{a}}{1+\frac{b^2}{a^2}}\right]$$

$$a \times \left[\frac{\frac{a^2-b^2}{a^2}}{\frac{a^2+b^2}{a^2}}\right] + b \times \left[\frac{\frac{2b}{a}}{\frac{a^2+b^2}{a^2}}\right]$$

$$= a\left[\frac{a^2-b^2}{a^2+b^2}\right] + b\left[\frac{2ab}{a^2+b^2}\right]$$

$$= \frac{a(a^2-b^2)+2ab^2}{a^2+b^2}$$

$$\frac{a^3-ab^2+2ab^2}{a^2+b^2}$$

$$\frac{a^3+ab^2}{a^2+b^2} = \frac{a[a^2+b^2]}{[a^2+b^2]} = a \text{ Ans}$$

(3) $\frac{\sec 8A - 1}{\sec 4A - 1} = \tan 8A \cot 2a$

$\sec\theta = \frac{1}{\cos\theta}$

$$\frac{\frac{1}{\cos 8A} - 1}{\frac{1}{\cos 4A} - 1} = \frac{\frac{1-\cos 8A}{\cos 8A}}{\frac{1-\cos 4A}{\cos 4A}}$$

$$\frac{1-\cos 8A}{\cos 8A} \times \frac{\cos 4A}{1-\cos 4A}$$

$1-\cos 2\theta \Rightarrow 2\sin^2$

$$\frac{1-\cos 2\cdot 4A}{\cos 8A} \times \frac{\cos 4A}{1-\cos 2\cdot 2A}$$

$$\frac{2\sin^2(4A)}{\cos 8A} \times \frac{\cos 4A}{2\sin^2(2A)}$$

$\sin 2\theta = 2\sin\theta \cdot \cos\theta$

$$\frac{2 \times \sin 4A \times \cos 4A \times \sin 4A}{\cos 8A \times 2\sin^2 2A}$$

$$\frac{\sin 8A \times \sin 4A}{\cos 8A \times 2\sin^2 2A}$$

$$\frac{\sin 8A \times 2\sin 2A \cdot \cos 2A}{\cos 8A \times 2 \cdot \sin 2A \cdot \sin 2A}$$

$\tan 8A \cdot \cot 2a = \tan 8A \cot 2a$

Hence Prove

(1) $\cos A = \frac{15}{17}$ {find the value of $\cos 2A$ if}

$$\cos 2A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

$\cos A = \frac{15}{17}$

$2\left(\frac{15}{17}\right)^2 - 1$

$2 \times \frac{225}{289} - 1$

$\frac{450}{289} - 1$

$\frac{450 - 289}{289}$

$\frac{161}{289}$ Ans

(2) $\sin A = \frac{4}{5}$ find the value of $\cos 2A$

$\cos 2A = 1 - 2\sin^2 A$

$= 1 - 2\left(\frac{4}{5}\right)^2$

$1 - 2 \times \frac{16}{25}$

$1 - \frac{32}{25}$

$\frac{25 - 32}{25}$

$-\frac{7}{25}$ Ans